

Two-Level Dual-View Event Relation Graphs for Sentence-Level Media Bias Detection

Mid-Project Submission Report

Anonymous ACL submission

Abstract

This report presents our mid-project progress on developing a two-level dual-view event relation graph architecture for sentence-level media bias detection. Building on Lei & Huang (2024)’s event relation graph approach, we introduce explicit separation of factual and interpretive events processed through hierarchical paragraph-level and document-level graphs. We have completed dataset preprocessing, event extractor training on MAVEN-ERE, event graph construction on BASIL and BiasedSents datasets, and factual/interpretive classification of all events. The remaining work involves implementing the two-level dual-view graph neural network architecture and conducting comprehensive evaluation. Our approach aims to improve bias detection by explicitly modeling how objective facts are framed through their connections to subjective characterizations.

1 Project Idea and Motivation/Objectives

1.1 Motivation

Media outlets increasingly exhibit partisan polarization, making it critical to detect ideological bias at a fine-grained level. While most prior work detects bias at the article or source level, bias often manifests in specific sentences that appear neutral but subtly frame facts to influence readers. The challenge is that bias sentences frequently seem factual when examined in isolation, requiring broader contextual understanding to reveal their persuasive intent.

A key insight is that events in bias sentences must be understood in association with other events in the document. For instance, describing historical events immediately after a statement can cast doubt on the statement’s credibility without explicit commentary. Current approaches process all events uniformly, but we observe that the relationship between factual events and interpretive characterizations is particularly revealing of bias.

1.2 Project Objectives

This project aims to improve sentence-level media bias detection by:

Primary Objective: Develop a two-level dual-view event relation graph that explicitly separates factual events from interpretive events and models their interactions at both paragraph and document levels.

Specific Goals:

- Improve precision by distinguishing between legitimate factual reporting and factual events used to frame interpretive claims
- Improve recall by capturing cross-paragraph narrative patterns where bias emerges from accumulated framing across the document
- Provide interpretable detection through explicit reasoning about event relationships and factual-interpretive distinctions

Innovation: Unlike Lei & Huang (2024) which processes all events in a flat unified graph, we introduce hierarchical processing with dual views, allowing the model to reason about how factual events are being framed through their connections to interpretive characterizations.

2 Background and Related Work

2.1 Media Bias Detection

Media bias research has evolved from source-level analysis to increasingly fine-grained detection. Early work by Groseclose and Milyo (2005) and Gentzkow and Shapiro (2010) focused on measuring outlet-level slant through citation patterns or word usage statistics. Baly et al. (2020) advanced article-level detection by combining textual features with source credibility signals.

The shift toward sentence-level analysis began with Fan et al. (2019), who created the BASIL dataset and demonstrated that individual sentences

serve different rhetorical purposes within articles. Lei et al. (2022) incorporated discourse structures, showing that sentence-level relations like elaboration and contrast help identify bias, achieving F1 of 49.10 on BASIL. However, their approach operates at the sentence level, potentially missing finer event-level patterns.

2.2 Event Relation Graphs

Event understanding has advanced significantly with large-scale datasets like MAVEN-ERE (Wang et al., 2022), which provides unified annotations for four fundamental event relations: coreference (whether events refer to the same occurrence), temporal (chronological ordering), causal (cause-effect relationships), and subevent (containment). These relations are trained jointly rather than in isolation, as they interact to form coherent narrative structures.

Lei & Huang (2024) pioneered applying event relation graphs to bias detection, demonstrating that understanding events through their relationships improves both precision and recall. Their relation-aware graph attention network processes heterogeneous event relations through type-specific attention mechanisms. However, they treat all events uniformly in a flat graph structure.

2.3 Graph Neural Networks

Graph Attention Networks (Veličković et al., 2018) enable learning adaptive importance weights for different neighbors. Relation-aware variants incorporate edge type semantics by learning separate transformation matrices for each relation type, which is essential for heterogeneous graphs like event structures. Hierarchical graph architectures that process information at multiple scales have shown success in document understanding tasks.

2.4 Research Gap

Current event-based bias detection processes all events uniformly, but bias often emerges specifically from the juxtaposition of factual events with interpretive characterizations. For example, reporting that someone "voted" (factual) is neutral, but connecting it to "undermined democracy" (interpretive) reveals ideological framing. Existing methods cannot explicitly reason about this factual-interpretive distinction.

3 Dataset Details

3.1 MAVEN-ERE (Event Extractor Training)

Purpose: Training event identification and relation extraction models

Source: Wang et al. (2022) - unified large-scale dataset for event relation extraction

Size:

- Training: 2,913 news articles
- Validation: 710 articles
- Test: 866 articles
- We use 20% of train/dev for computational efficiency

Annotations: Every article contains:

- Event mentions with trigger words and text spans
- Four relation types exhaustively annotated between all event pairs
- Coreference chains grouping mentions of the same event
- Temporal ordering (before, after, overlap relations)
- Causal structures (causes, caused-by relations)
- Subevent hierarchies (contains, contained-by relations)

Domain: General news covering politics, business, disasters, sports

Quality: Professional annotations with inter-annotator agreement validation

3.2 BASIL (Bias Detection)

Purpose: Primary evaluation dataset for sentence-level bias detection

Source: Fan et al. (2019)

Size:

- 300 articles covering 100 political events (2010-2019)
- 3 articles per event from NYT, Huffington Post, Fox News
- 7,977 total sentences
- 1,623 bias sentences (20.34% positive class)

Annotation Types:

- **Lexical bias:** Overt ideological language (e.g., "slammed", "celebrated")
- **Informational bias:** Selective inclusion/omission of context
- We combine both types as a unified bias class

Article Structure: Each triplet covers the same event from three outlets with different political leanings (left, center, right), enabling comparative analysis of how different sources frame identical events.

Label Distribution: Imbalanced dataset with roughly 80% non-bias sentences, requiring careful handling during training.

3.3 BiasedSents (Secondary Evaluation)

Purpose: Independent validation dataset with different annotation scheme

Source: Lim et al. (2020)

Size:

- 46 articles (2017-2018)
- 842 sentences
- 290 bias sentences (34.44% positive class)

Annotations:

- Original: 4-point scale (not biased, slightly biased, biased, very biased)
- Our processing: Binarize into non-bias (first two) and bias (latter two)
- 5 independent annotators with majority voting for ground truth

Difference from BASIL: Higher bias ratio (34% vs 20%), different time period, different annotation methodology provides useful test of generalization.

4 Proposed Methodology

4.1 Overall Architecture

Our approach consists of a six-stage pipeline combining pre-trained event extractors with a novel two-level dual-view graph neural network. Figure 1 illustrates the complete architecture.

Pipeline Overview:

1. Preprocess MAVEN-ERE from JSONL format to individual article JSONs

2. Train joint event extraction model on MAVEN-ERE
3. Apply extractors to BASIL and BiasedSents to build event graphs
4. Classify each extracted event as factual or interpretive using Gemini API
5. Construct two-level dual-view graphs separating factual/interpretive events
6. Train hierarchical GNN for bias detection

4.2 Event Relation Graph Foundation

Our event relation graphs represent document narratives through events as nodes connected by four relation types. Each event corresponds to a verb or action (e.g., "voted", "announced", "warned") with associated arguments identifying participants and objects.

Relation Types:

Coreference identifies when multiple mentions refer to the same underlying event occurrence. For example, "the vote" and "the decision" may be coreferent. This is crucial for bias detection because authors may use different framings for the same event.

Temporal relations establish chronological ordering with three subtypes: before (Event A precedes Event B), after (Event A follows Event B), and overlap (simultaneous occurrence). Temporal context often reveals bias through strategic sequencing of events to suggest causality or pattern.

Causal relations explicitly represent cause-effect structures with two directional subtypes: causes (Event A led to Event B) and caused-by (Event A resulted from Event B). Bias frequently appears in claimed causality that lacks supporting evidence.

Subevent relations capture hierarchical event structures with two subtypes: contains (Event A encompasses Event B) and contained-by (Event A is part of Event B). This helps identify when specific details are emphasized or omitted to support a narrative.

4.3 Factual vs Interpretive Event Distinction

The core innovation of our approach is explicitly separating events by their epistemological status:

Factual Events are objectively verifiable occurrences that can be confirmed through official

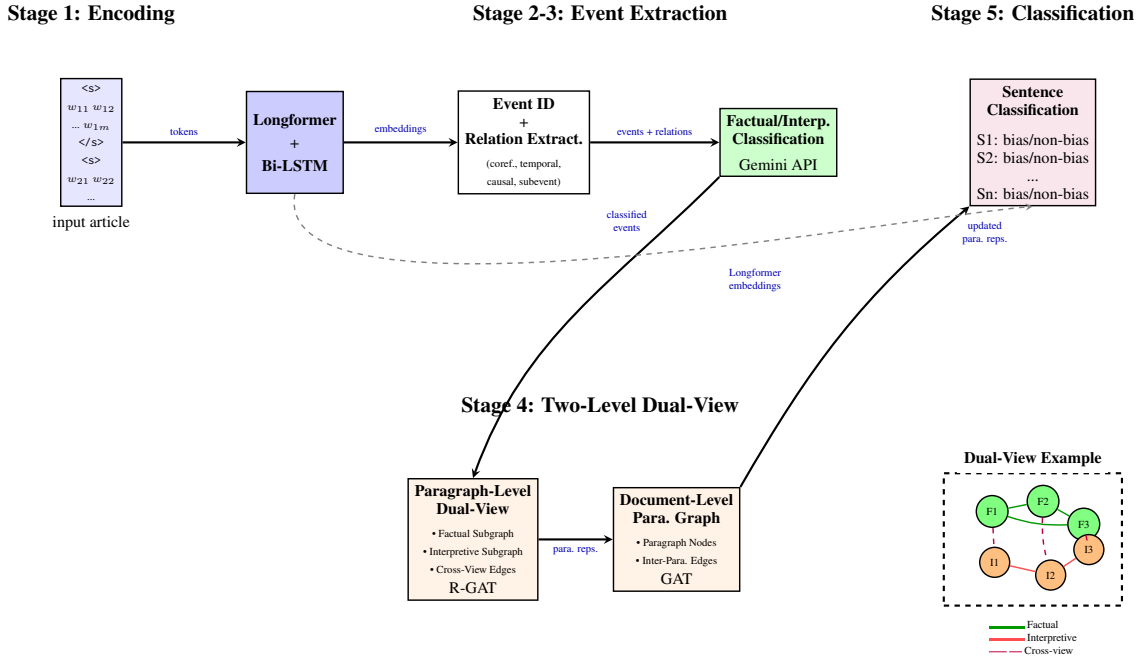


Figure 1: Architecture of our two-level dual-view event relation graph approach for sentence-level media bias detection. The pipeline processes articles through five stages: encoding with Longformer and BiLSTM, event and relation extraction, factual/interpretive classification, two-level dual-view graph processing (paragraph-level then document-level), and final sentence classification combining graph representations with base embeddings.

records, measurements, or observable evidence. These include physical actions like "traveled" or "met", official proceedings like "voted" or "ruled", measurable changes like "increased by 10%" or "reached \$5 billion", and formal declarations like "announced" or "published". Factual events form the informational backbone of news reporting.

Interpretive Events involve subjective characterization, evaluation, or speculation that cannot be directly verified. These include opinion attributions like "critics claimed" or "experts warned", modal expressions like "could harm" or "might benefit", evaluative characterizations like "slammed" or "celebrated", and cognitive states like "worried" or "hoped". Interpretive events reveal how journalists frame factual occurrences.

Key Distinction: The verb "announced" is factual because an announcement objectively occurred, while "warned" is interpretive because it implies negative evaluation. Similarly, "increased to 8%" is factual (measurable), but "soared to 8%" is interpretive (the term "soared" carries evaluative judgment).

Classification Implementation: We use Gemini 2.5 Flash Lite API with carefully engineered prompts that provide clear definitions and examples of each category. For each event, we provide

the sentence context to help the model understand usage. The API returns a classification label, confidence score, and brief reasoning explanation.

4.4 Two-Level Hierarchical Structure

Based on advisor feedback, we implement a two-level hierarchy rather than three levels:

Rationale for Two Levels: Modern pre-trained language models like Longformer already effectively capture relationships between events within a single sentence through self-attention mechanisms. Therefore, sentence-level graph processing is redundant. We focus computational resources on paragraph-level and document-level processing where additional structure provides unique value.

Level 1 - Paragraph Graphs: Capture local cross-sentence event patterns within coherent discourse units. Paragraphs typically contain related sentences discussing sub-topics, making them natural units for localized event analysis.

Level 2 - Document Graph: Captures global cross-paragraph narrative structure. Documents contain multiple paragraphs that together construct the overall narrative arc, and bias often emerges from how these segments connect.

4.5 Dual-View Graph Structure

At the paragraph level, we construct three types of substructures:

Factual Event Subgraph: Contains only events classified as factual, connected by the four relation types (coreference, temporal, causal, subevent). This subgraph represents the objective informational structure of the paragraph.

Interpretive Event Subgraph: Contains only interpretive events, also connected by the four relation types. This subgraph captures the evaluative and characterization structure overlaid on the facts.

Cross-View Edges: Bidirectional connections between factual and interpretive events that co-occur or relate to each other. These edges are labeled as "interprets" (factual to interpretive direction) or "supported-by" (interpretive to factual direction). Cross-view edges are created when events appear in the same sentence, share argument overlap, or have any extracted relation between them.

Why Dual-View Matters: Bias frequently emerges not from interpretive events alone, but from how interpretive characterizations are anchored to factual events. Separating the views allows explicit modeling of this factual-interpretive interaction.

4.6 Paragraph-Level Processing

For each paragraph, we process the three subgraphs through specialized graph neural network modules:

Factual Subgraph Processing: A relation-aware graph attention network updates each factual event's embedding by aggregating information from connected factual events. The network learns separate attention weights for each of the four relation types, allowing different relations to contribute differently to the representation.

Interpretive Subgraph Processing: An analogous relation-aware GAT processes interpretive events, learning how interpretive characterizations relate to each other through the same four relation types.

Cross-View Attention: After within-view processing, we apply cross-view attention that allows factual event representations to be influenced by connected interpretive events, and vice versa. This bidirectional mechanism captures how facts are being framed.

Paragraph Representation: We aggregate all updated factual event embeddings into a single factual summary vector, aggregate all updated inter-

pretive event embeddings into an interpretive summary vector, and combine these through a learned transformation to produce a final paragraph representation that encodes both the factual content and its interpretive framing.

4.7 Document-Level Processing

Paragraph as Super-Nodes: Each paragraph becomes a single node in the document graph, represented by its 768-dimensional paragraph vector from Level 1 processing.

Paragraph Connections: We create edges between paragraphs based on:

- Sequential adjacency (paragraph N connects to paragraph N+1)
- Coreference chains (paragraphs containing coreferent events are connected)
- Semantic relatedness (paragraphs discussing related entities or sub-topics)

Document-Level GAT: A standard graph attention network processes the paragraph graph, allowing paragraph representations to be updated with document-wide context. This captures how early paragraphs set up context for later ones, and how conclusions relate back to earlier content.

4.8 Sentence Classification

For final bias prediction, each sentence receives three sources of information:

Longformer Encoding: The base contextualized representation from the pre-trained language model captures lexical patterns and immediate context.

Paragraph Context: The updated paragraph representation from document-level processing provides information about how the paragraph fits into the broader narrative and whether it contains factual-interpretive framing patterns.

Event Aggregation: The mean of all event embeddings appearing in the sentence provides direct signal about which events are being reported and how they relate to others.

These three representations are concatenated and fed through a classification MLP to produce the final bias/non-bias prediction. The model is trained end-to-end using cross-entropy loss on the sentence-level ground truth labels.

4.9 Differences from Lei & Huang (2024) Baseline

Our approach diverges from the baseline in four fundamental ways:

Dual-View Separation: We separate factual and interpretive events into distinct subgraphs rather than processing all events uniformly, enabling explicit modeling of fact-framing patterns.

Hierarchical Processing: We use two levels (paragraph and document) instead of flat processing, capturing both local and global narrative structure.

No Knowledge Distillation: Lei & Huang train auxiliary heads to mimic event extractor outputs (soft label distillation). We simplify by directly using pre-extracted events and relations, training only the bias classifier.

Full Supervision: We train exclusively on sentence-level bias labels rather than combining them with event/relation prediction losses, focusing the model entirely on the target task.

5 Evaluation Plan

5.1 Evaluation Metrics

Following established practice in sentence-level bias detection:

- Precision of bias class (how many predicted bias sentences are actually biased)
- Recall of bias class (how many true bias sentences we successfully identify)
- F1 score of bias class (harmonic mean, our main evaluation criterion)

5.2 Experimental Protocol

Cross-Validation Strategy: Ten-fold cross-validation with fold division at the article level to prevent information leakage. Sentences from the same article never appear in both training and testing folds.

Fold Configuration: Each iteration uses 8 folds for training, 1 fold for validation (early stopping and hyperparameter selection), and 1 fold for testing. We rotate through all 10 folds to obtain predictions for every article.

BASIL-Specific: Folds are constructed to maintain balanced representation from the three news outlets (NYT, Huffington Post, Fox News) in each fold.

Training Details:

- BASIL: 5 epochs maximum per fold
- BiasedSents: 10 epochs maximum (smaller dataset requires more epochs)
- Early stopping based on validation F1
- AdamW optimizer with linear warmup scheduling

Final Evaluation: Average metrics across all 10 test folds, report standard deviations to indicate variance.

5.3 Ablation Studies

To understand which components contribute to performance:

Dual-View Ablation: Merge factual and interpretive events into a single unified graph (removes view separation while keeping hierarchy).

Hierarchy Ablation: Process all events in a flat graph without paragraph or document levels (removes hierarchy while potentially keeping dual-view).

Cross-View Edge Ablation: Remove edges connecting factual to interpretive events (keeps separate views but no interaction).

6 Progress So Far

6.1 Completed Objectives (Weeks 1-3)

6.1.1 Dataset Preparation - COMPLETED

What We Did: Converted MAVEN-ERE dataset from its original JSONL format (one massive file with each line containing a complete article) into individual JSON files compatible with the event extraction framework.

Key Accomplishments:

- Processed 3,623 articles into structured individual files
- Tokenized articles using consistent tokenization scheme
- Extracted event mentions with trigger words, spans, and identifiers
- Generated all possible event pairs maintaining natural textual order
- Converted temporal relation annotations (consolidated SIMULTANEOUS, OVERLAP, BEGINS-ON, ENDS-ON, CONTAINS into single "overlap" category)

487	• Adjusted relation directionality to maintain natural textual order	529
488		
489	• Created standardized format with event labels and relation labels for all pairs	530
490		
491	Output: Organized dataset with train directory (2,913 files) and dev directory (710 files) ready for model training.	531
492		
493		
494	6.1.2 Event Extractor Training - COMPLETED	532
495		533
496	What We Did: Trained a joint neural model on MAVEN-ERE to simultaneously learn event identification and relation classification on MAVEN-ERE.	534
497		535
498		536
499	Architecture Implemented:	537
500		538
501	• Longformer-base-4096 as document encoder (handles long articles up to 4096 tokens)	539
502		
503	• BiLSTM layer for additional contextualization	
504		
505	• Event identification head predicting event vs non-event for each token	
506		
507	• Four relation classification heads for coreference, temporal, causal, and subevent	
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509	Training Configuration:	
510	• Used 20% of MAVEN-ERE (583 training articles, 142 validation articles) for computational efficiency	
511		
512		
513	• Joint training framework where all five components train simultaneously	
514		
515	• 5 training epochs with intermediate validation checkpoints	
516		
517	• Batch size 1 (process one article at a time due to variable length)	
518		
519	• Different learning rates: Longformer layers at 1e-5, new classification layers at 1e-4	
520		
521	• Weighted cross-entropy loss to handle severe class imbalance in relations	
522		
523	Performance Achieved:	
524	• Event Identification: Macro F1 = 89.40 (precision 87.31, recall 91.81)	
525		
526	• Coreference Resolution: CoNLL F1 = 88.30 (average of MUC, B ³ , CEAF _e , BLANC metrics)	
527		
528		
	• Temporal Classification: Macro F1 = 47.04	529
	• Causal Classification: Macro F1 = 56.01	530
	• Subevent Classification: Macro F1 = 46.21	531
	Validation: The strong event identification performance (F1 89.40) ensures high-quality event detection for downstream bias detection. Relation extraction F1 scores of 47-56 are competitive with state-of-art on this challenging task.	532
	Saved Checkpoints: Six trained models preserving best performance for each component plus overall best checkpoint.	533
		534
	6.1.3 Event Graph Extraction on Target Datasets - COMPLETED	535
		536
	What We Did: Applied trained extractors to BASIL and BiasedSents datasets to identify events and extract their relationships.	537
		538
	Process:	539
	• Tokenized each article using Longformer tokenizer with special tokens	540
		541
	• Applied event identification model to predict event probability for every word	542
		543
	• Identified event triggers using 0.5 probability threshold	544
		545
	• Created all possible pairs from identified events	546
		547
	• Applied four relation classifiers to predict relation probabilities for each pair	548
		549
	• Stored both probabilistic predictions and hard labels	550
		551
	Coverage:	552
	• BASIL: 300 articles processed, approximately 15-20 events per article	553
		554
	• BiasedSents: 46 articles processed, approximately 10-15 events per article	555
		556
	Output Structure: Enhanced article JSONs containing original metadata, sentence-level bias labels, token-level event probabilities, identified event tokens, and complete event pair predictions with probabilities for all four relation types.	557
	Validation: Manual inspection of sample outputs confirmed appropriate event identification and meaningful relation extraction.	558
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6.1.4 Factual/Interpretive Classification - COMPLETED

What We Did: Developed and executed automated classification system using Gemini API to label every extracted event as factual or interpretive.

Prompt Engineering: Designed comprehensive prompts with clear definitions distinguishing objectively verifiable occurrences from subjective characterizations, multiple examples covering diverse verb types, explicit guidance on ambiguous cases, and structured JSON output requirements.

Execution Results:

- Successfully classified all events in 346 articles (300 BASIL + 46 BiasedSents)
- Each event now has factual/interpretive label, confidence score, and reasoning explanation
- Observed distribution approximately 60-70% factual, 30-40% interpretive events
- Created complete classified versions of all event graph files

6.2 Planned Work (Weeks 4-7)

6.2.1 Two-Level Dual-View Model Implementation - PLANNED (Weeks 4-5)

Timeline: October 3 - October 16 (2 weeks)

Major Implementation Components:

Data Loading System (Week 4): We will develop a custom data loader that reads the classified event graphs and organizes them for hierarchical dual-view processing. The loader will detect paragraph boundaries using sentence grouping heuristics or topic shift patterns, separate events into factual and interpretive categories based on their classification labels, and create paragraph-indexed data structures. Each batch will include paragraph identifiers mapping sentences to paragraphs and view indices indicating which events are factual versus interpretive.

Paragraph-Level Dual-View Graph Neural Network (Weeks 4-5): We will implement a relation-aware graph attention network with dual processing pathways. The factual pathway will process factual events using type-specific attention weights for each of the four relation types (coreference, temporal, causal, subevent). A parallel interpretive pathway will process interpretive events with its own learned parameters. We will design a cross-view attention mechanism allowing

bidirectional information flow between factual and interpretive event representations. The paragraph aggregation module will combine the factual event summary and interpretive event summary into a unified paragraph representation. Special handling will address edge cases such as paragraphs containing only one view type.

Document-Level Graph Neural Network (Week 5): We will implement a standard graph attention network operating on paragraph-level representations as super-nodes. The module will create connections between paragraphs based on sequential adjacency and coreference chains linking events across paragraphs. The aggregation mechanism will propagate document-wide contextual information to update each paragraph's representation.

Main Model Architecture (Week 5): We will build the complete end-to-end model integrating all components. The Longformer encoder will generate base token embeddings, which flow into the paragraph-level dual-view processor. We will implement the logic that organizes events by paragraph and separates them by view type, builds the dual-view graphs for each paragraph, and passes paragraph representations to the document-level processor. The updated paragraph representations will be mapped back to their constituent sentences. The final classification layer will combine three information sources: Longformer sentence embeddings, updated paragraph representations, and aggregated event embeddings from the sentence. Unlike the baseline approach, we will not include knowledge distillation losses, focusing solely on the bias classification objective.

Supporting Functions (Week 5): We will develop utility functions for paragraph boundary detection from sentence sequences, event filtering by paragraph identifier and view type, relation extraction scoped to event pairs within the same paragraph, cross-view edge identification based on sentence co-occurrence and argument overlap, and paragraph graph construction creating edges between related paragraph nodes.

Implementation Scope: We estimate approximately 700-900 lines of neural network architecture and data processing code for each dataset variant (BASIL and BiasedSents require separate implementations due to different data formats).

6.2.2 Training and Evaluation - PLANNED (Weeks 6-7)

Timeline: October 17 - October 30

Training Procedure: We will execute 10-fold cross-validation on both datasets. For BASIL, each fold will train for a maximum of 5 epochs with early stopping triggered when validation F1 stops improving. For BiasedSents, we will use 10 epochs per fold to account for the smaller dataset size. Each fold will save its best model checkpoint based on validation performance. Throughout training, we will monitor loss curves and validation metrics to detect overfitting or convergence issues requiring hyperparameter adjustment.

Evaluation Procedure: After completing all 10 folds, we will collect predictions for every article’s test fold and calculate precision, recall, and F1 score for the bias class. We will compute macro-averaged metrics across both classes and generate detailed classification reports with confusion matrices. Statistical significance testing using paired t-tests or bootstrap resampling will compare our approach against baselines across the 10 folds.

Baseline Experiments: We will implement and train three baseline systems for comparison. The Longformer baseline will use the same encoder and BiLSTM but without any event graph processing. We will implement the Lei and Huang (2024) approach with flat single-view event graphs. A simple feature concatenation baseline will append event probability vectors to sentence embeddings without graph attention. All baselines will train on identical cross-validation folds to ensure fair comparison.

Ablation Studies: We will conduct systematic ablation experiments to isolate the contribution of each architectural component. The dual-view ablation will merge factual and interpretive events into a unified graph while maintaining hierarchical processing. The hierarchy ablation will process all events in a flat graph without paragraph or document levels. The cross-view edge ablation will maintain separate factual and interpretive sub-graphs but remove all connections between them. Relation type ablations will systematically exclude each of the four relation types individually. Each ablation will run complete 10-fold cross-validation for statistically valid comparison.

Analysis Activities: We will conduct comprehensive error analysis examining false positive and false negative predictions to understand failure modes. Attention weight visualization will reveal

which relation types and cross-view connections receive highest importance during classification. We will analyze the distribution of factual versus interpretive events in bias versus non-bias sentences to validate our core hypothesis. For BASIL, we will break down performance by news outlet (NYT, Huffington Post, Fox News) to understand if certain sources exhibit different bias patterns. We will also analyze which bias types (lexical versus informational) benefit most from our dual-view approach.

7 Tentative Timeline

7.1 Revised Timeline (Adjusted for November 1 Deadline)

Weeks 1-2 (Sep 18 - Oct 1): COMPLETED

- MAVEN-ERE preprocessing completed
- Event extractor training completed
- Event graph construction completed
- Factual/interpretive classifier development completed

Week 3 (Oct 2): COMPLETED

- Executed factual/interpretive classification on all 346 articles
- Quality validation confirmed appropriate classifications
- Distribution analysis completed
- All classified data ready for model implementation

Week 4 (Oct 3 - Oct 9): CURRENT WEEK

- Implement paragraph boundary detection module
- Implement modified data loading system for both datasets
- Begin paragraph-level dual-view GNN implementation
- Implement relation-aware attention mechanism core

Week 5 (Oct 10 - Oct 16):

- Complete paragraph-level GNN with cross-view attention

758	• Implement document-level GNN module
759	• Integrate all components into main model
760	class
761	• Unit testing and debugging of each module
762	• Integration testing on small data subset
763	Week 6 (Oct 17 - Oct 23):
764	• Launch 10-fold cross-validation training on
765	BASIL
766	• Launch 10-fold cross-validation training on
767	BiasedSents
768	• Monitor training progress and adjust hyperpa-
769	rameters
770	• Run baseline comparison experiments
771	• Begin ablation studies on completed folds
772	Week 7 (Oct 24 - Oct 30):
773	• Complete all training runs and ablation exper-
774	iments
775	• Compile results tables and visualizations
776	• Error analysis and interpretation
777	• Begin final report writing
778	Final Days (Oct 31 - Nov 1):
779	• Finalize report and results
780	• Final submission preparation
781	• Submit by November 1 deadline
782	7.2 Critical Path Items
783	Must Complete by Oct 16: Full model implemen-
784	tation and successful test run
785	Must Complete by Oct 30: All training and
786	primary experiments finished
787	Must Complete by Nov 1: Final report submit-
788	ted
789	7.3 Contingency Plans
790	If implementation takes longer than expected:
791	• Prioritize paragraph-level only (skip
792	document-level if necessary)
793	• Reduce number of ablations to essential ones
794	• Train on BASIL only if time extremely con-
795	strained (primary dataset)

8 Expected Contributions	796
8.1 Methodological Contributions	797
Novel Architecture: First event relation graph	798
approach explicitly separating factual and interpre-	799
tive events with dedicated processing pathways and	800
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Hierarchical Event Processing: Demonstration	802
that paragraph-level and document-level structures	803
capture different bias manifestation patterns.	804
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rect usage of pre-extracted events without knowl-	806
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Huang 2024: F1=52.00 on BASIL).	812
Ablation Insights: Understanding whether dual-	813
view separation provides value, which hierarchical	814
level contributes more, which relation types are	815
most informative, and whether cross-view edges	816
are essential.	817
Event Analysis: Factual/interpretive distribu-	818
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tematic patterns across outlets or bias types.	820
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